PARSHWANATH CHARITABLE TRUST'S
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Department of Computer Science and Engineering
Data Science

Smart Privacy Shield:

Selective Encryption and Blockchain-based Image Security

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Project Guide

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Outline

- Introduction
- Literature Survey of the existing systems
- Limitations of the existing systems
- Problem statement
- System Design
- Technologies and methodologies
- Implementation
- Conclusion
- References

Sustainable Development Goals (SDG) mapped

- **SDG 9: Industry, Innovation, and Infrastructure**

  Developing secure digital infrastructure for private data handling.

- **SDG 16: Peace, Justice, and Strong Institutions**

  Protecting individual privacy and fundamental freedoms in a digital-first society.

Introduction

Real-time Observations of Problem Domain:

- Exponential growth in digital image storage on centralized cloud servers increasing the vulnerability of sensitive data.
- Stored assets often contain unmasked Personally Identifiable Information (PII) like faces and text, risking privacy in case of data breaches.
- Conventional full-image encryption in storage is visually "loud," flagging encrypted files for targeted attacks by hackers.
- Increasing concerns over lack of transparent access logs for sensitive images stored in centralized databases.

Motivation for Project:

- Necessity for a "hidden" security layer within storage environments that does not alert attackers to the existence of secret data.
- Desire to combine Computer Vision (OpenCV) with Blockchain to create a verifiable, private storage ecosystem.
- Need for decentralized ownership proof to prevent unauthorized image duplication and access within centralized storage.
- Ambition to allow users to store images with context preserved while keeping sensitive ROI (Region of Interest) cryptographically secure.

Introduction

Project Objectives:

- Design a system to identify sensitive regions (faces, text, license plates) using Computer Vision and Machine Learning.
- Implement a selective encryption mechanism protecting only confidential parts while preserving visual context.
- Ensure data integrity and transparency by recording image metadata, hashes, and access logs on a blockchain.
- Develop an NFT-based secure transmission method combining cryptography and steganography for authorized users only.

Literature Survey of the existing system

Sr. no	Title	Author(s)	Year	Methodology	Outcomes	Result
1	Secure Data Hiding using LSB Steganography	Johnson, K. et al.	2024	Embedding data into the Least Significant Bits of pixel values.	High data embedding capacity in carrier images.	Limitation: Susceptible to statistical steganalysis and detection.
2	Decentralized Media Ownership via Blockchain	Chen, L. & Zhang, H.	2025	Utilizing Ethereum smart contracts and IPFS for hash storage.	Provides an immutable and verifiable ownership ledger.	Limitation: High transaction costs (gas) for frequent transfers.

Literature Survey of the existing system

Sr. no	Title	Author(s)	Year	Methodology	Outcomes	Result
3	Selective ROI Encryption for Privacy	Kumar, S.	2025	Identifying regions of interest (ROI) like faces for targeted AES encryption.	Preserves background context while protecting PII.	Limitation: Highly dependent on the accuracy of ROI detection models.
4	Hybrid Cryptography and Steganography	Smith, J. & Lee, D.	2026	Dual-layer security: Encrypting data before steganographic embedding.	Significantly enhanced robustness against unauthorized access.	**Limitation:** Increased computational overhead and processing time.

Limitations of existing systems

Survey Analysis Observations

- **Fragmented Security:** Most current systems offer either invisibility (steganography) or data protection (encryption), but rarely integrate both.
- **Centralization Vulnerabilities:** Existing image lockers rely on central servers, presenting single points of failure and trust issues.
- **Detection Accuracy Risks:** Selective encryption performance is entirely dependent on the precision of ROI (Region of Interest) detection models.
- **Contextual Utility Loss:** Standard full-image encryption removes all visual context, making the file unusable for legitimate previewing.

The Unsolved Gap

- **Privacy-Utility Trade-off:** The problem remains unsolved as users currently must sacrifice image utility for total data privacy.
- **Absence of Ownership Proof:** Most steganographic tools lack a verifiable mechanism to prove digital ownership after sharing.

Limitations of existing systems

Methodologies Performing Better

- **Decentralized Ledgers:** Blockchain offers superior data integrity and transparency for recording access logs compared to standard databases.
- **ML-Driven Identification:** Automated detection of faces and license plates is performing better than manual masking techniques**.**

Justification & Confidence

- **Addressing Integration Gaps:** We are confident in contributing a solution that bridges the gap between digital invisibility and verifiable ownership.
- **Value Addition:** Our Smart Privacy Shield justifies the use of selective encryption and NFTs to solve the invisibility-utility-ownership conflict.

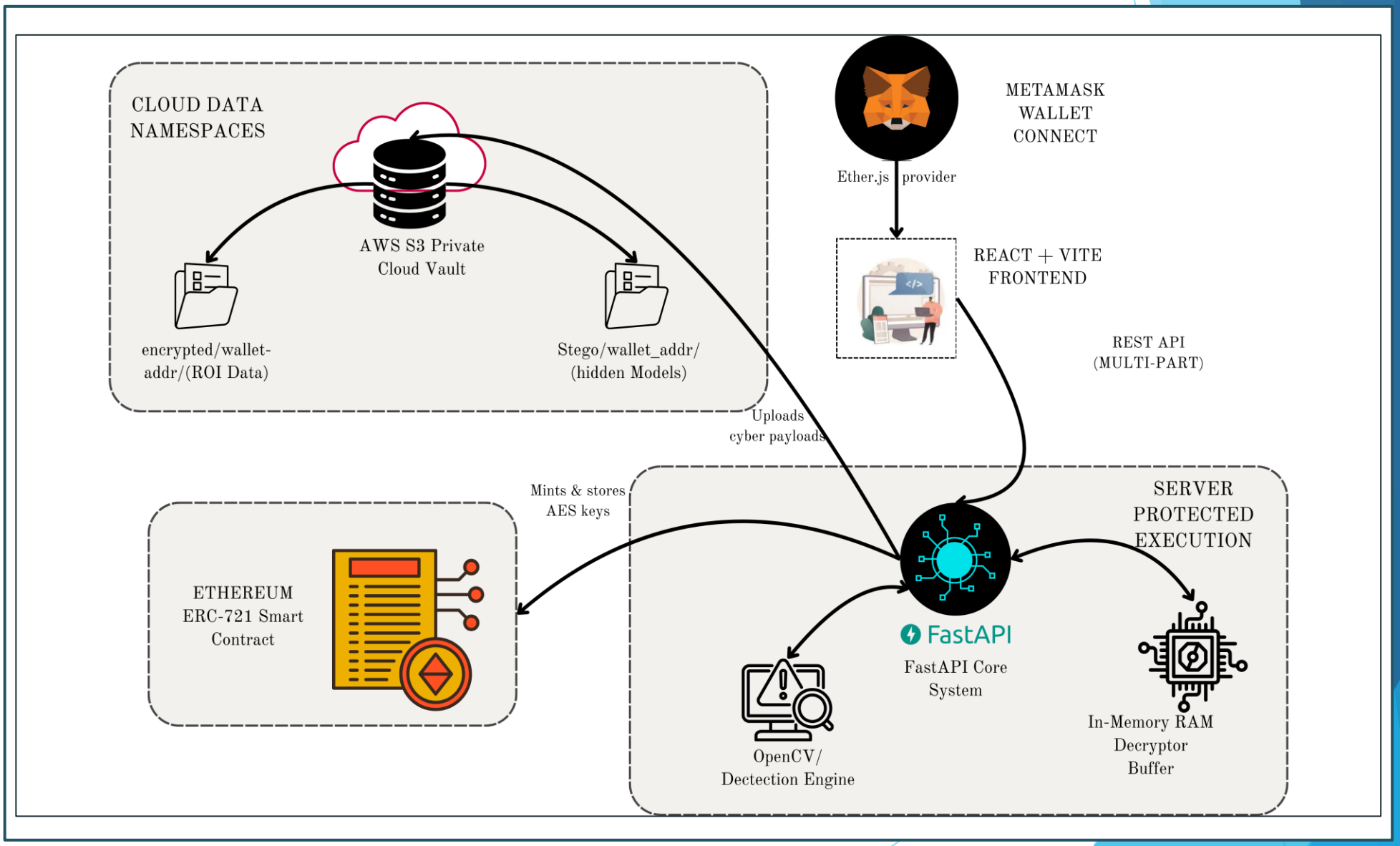
Problem statement

Developing a hybrid system that automates the identification of sensitive image regions and secures them through selective encryption and steganography, while establishing immutable digital ownership using blockchain technology.

Modern image sharing lacks a granular security mechanism that protects sensitive data without compromising the overall utility of the image or relying on centralized, untrusted servers.

The Smart Privacy Shield aims to solve this by integrating Computer Vision for selective encryption, Steganography for invisibility, and Blockchain for verifiable ownership.

System Design



Technologies and methodologies

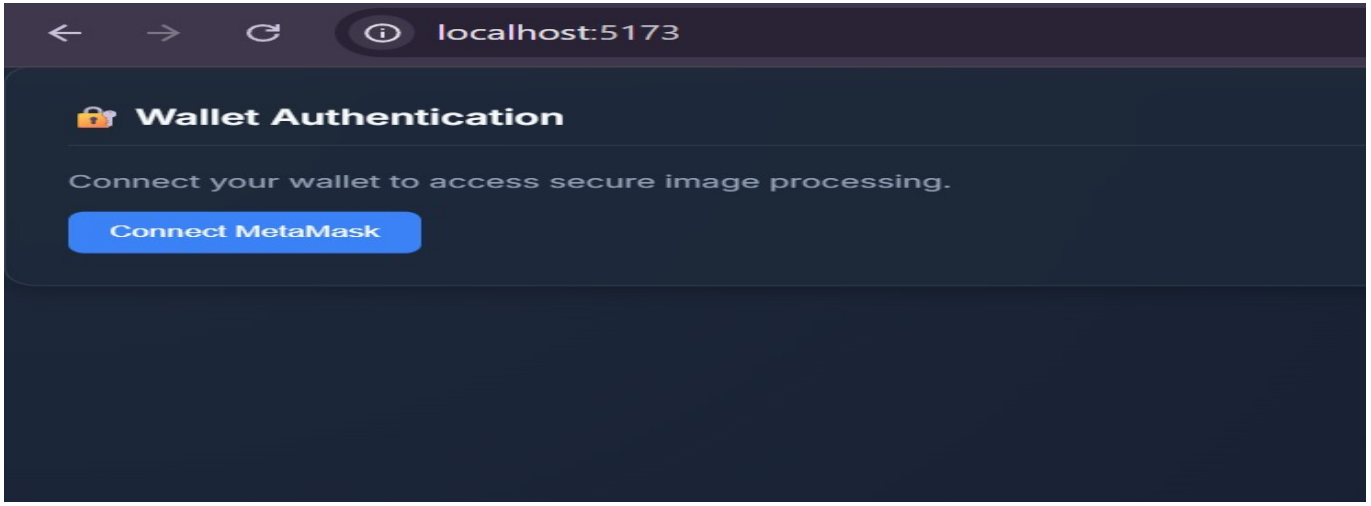
Core Tech Stack & Development Environment

- Python & Fast API: Utilized for the backend to handle high-performance API requests and real-time image processing.
- OpenCV Library: The primary tool for Computer Vision tasks, specifically for identifying and manipulating sensitive **Regions of Interest (ROI).**
- Solidity & Hardhat: Used for developing and testing Ethereum-compatible smart contracts that manage **decentralized ownership.**
- React : Employed to build a secure, application for localized encryption and user interaction.

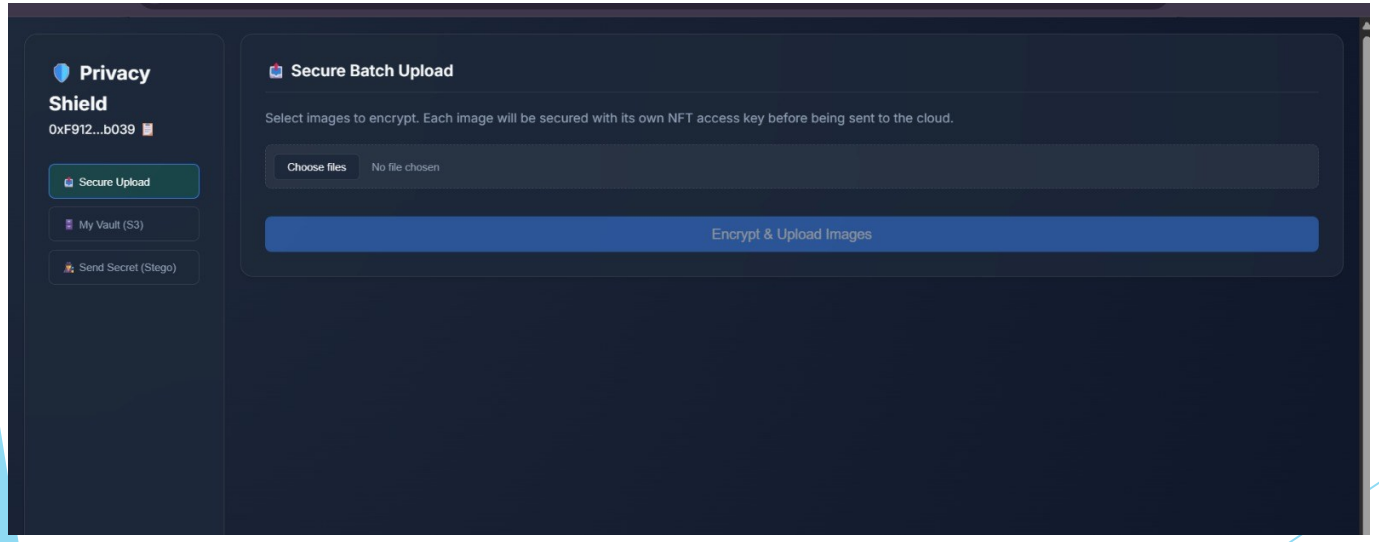
Techniques & Technical Details

- Sensitive ROI Detection (ML-Based): Employs Machine Learning models within the OpenCV framework to accurately pinpoint faces, text, and license plates.
- Selective Cryptography: Theory involves applying encryption algorithms solely to the confidential pixel blocks of an image while preserving the remaining visual context.
- Least Significant Bit (LSB) Steganography: A data-hiding method that substitutes the least significant bits of a pixel with encrypted data to ensure visual invisibility.
- Blockchain-Based Ledger: Uses a decentralized approach to record image metadata, cryptographic hashes, and immutable access logs.

Implementation

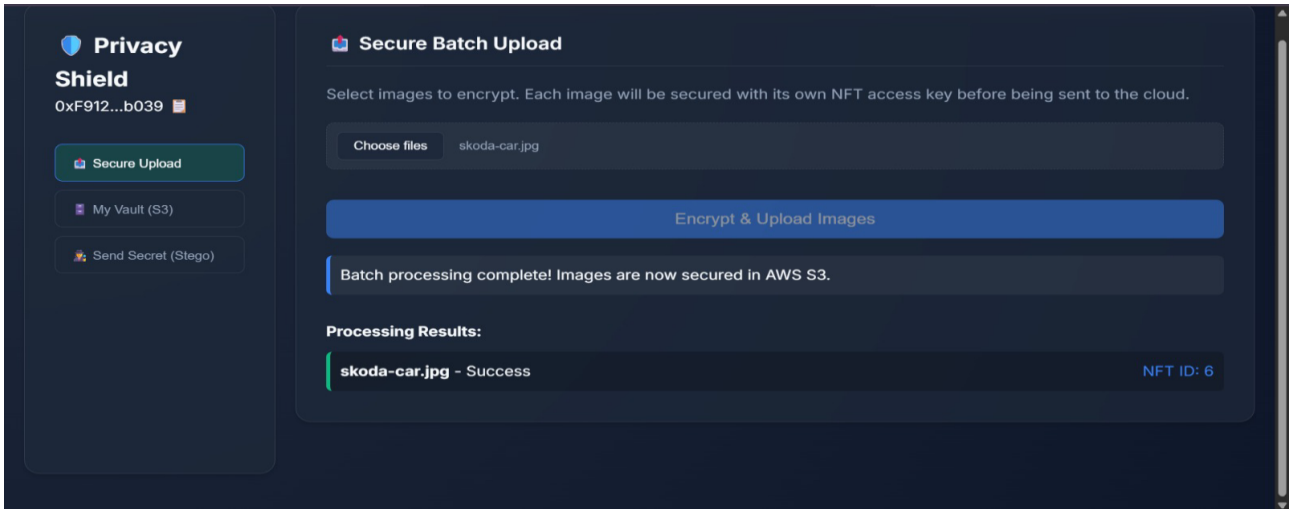


User login page

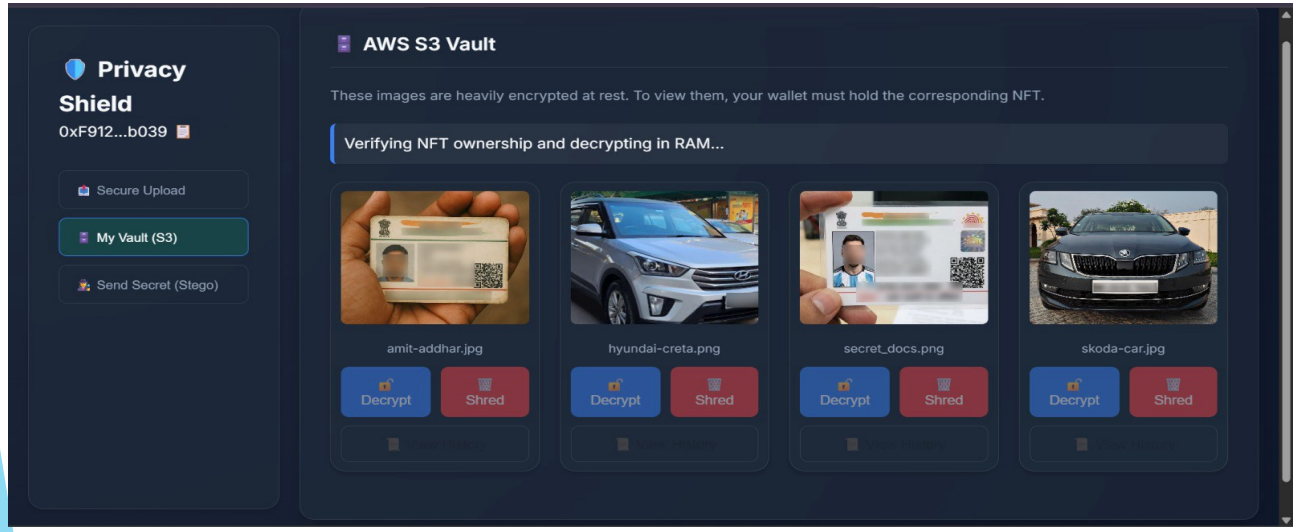


Dashboard & File Upload

Implementation

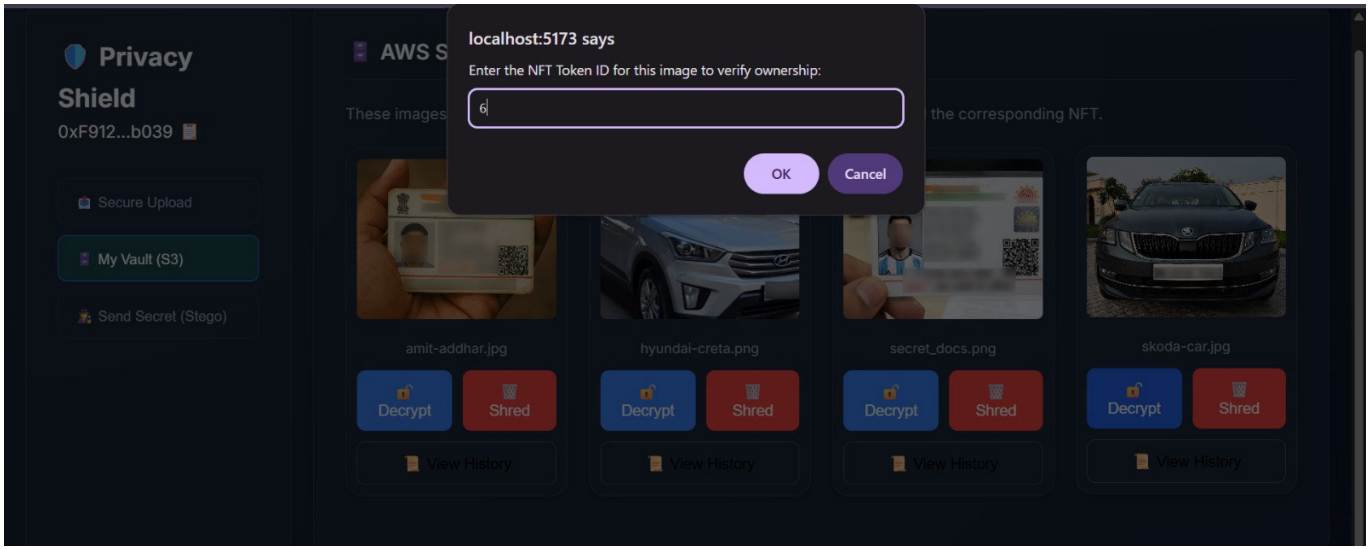


File Upload

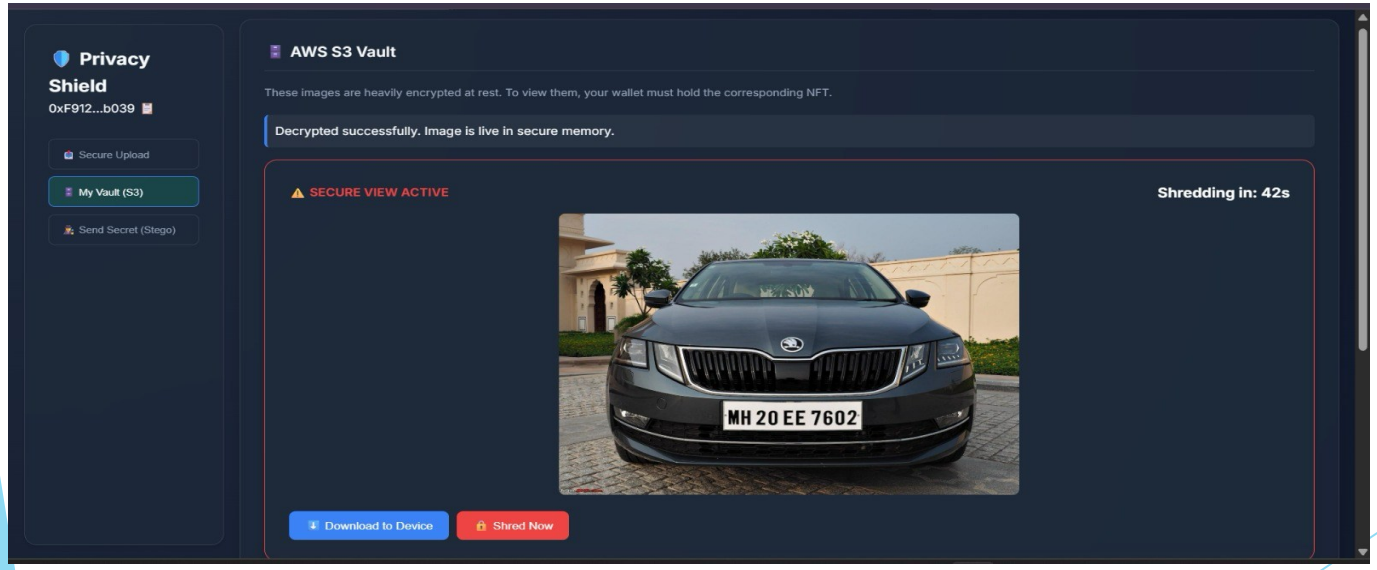


Vault

Implementation



File decryption



Decryption Viewer

Conclusion

- **Integration of Privacy & Utility:** The project successfully bridges the gap between digital invisibility and image utility through a hybrid security model
- **Effective ROI Protection:** Selective encryption successfully secures sensitive regions like faces and license plates while maintaining the overall context of the carrier image.
- **Decentralized Ownership:** The implementation of NFT-based logic and blockchain logs establishes a verifiable and immutable proof of digital ownership.
- **Steganographic Robustness:** By hiding encrypted data within carrier files, the system significantly reduces the visibility of sensitive information to potential attackers.
- **Fulfillment of Objectives:** All core objectives from automated detection to secure, authorized transmission have been addressed within the developed framework.

Future Scope

- **Scaling & Optimization:** Future enhancements will focus on improving detection accuracy for low-light images and optimizing blockchain gas efficiency for large-scale metadata.
- **Cross-Platform Expansion:** Expanding the current web application implementation into a mobile-first privacy suite for real-time social media protection.

References

- [1] Johnson, K., et al. "A Novel LSB Steganography Technique for Enhancing Cloud Security." Scientific Research Publishing, 2024.
- [2] Chen, L., & Zhang, H. "Design of a Decentralized Social Media Platform Using Blockchain to Enhance Privacy and User Trust." International Journal of Advanced Studies in Information Systems, 2025.
- [3] Kumar, S. "ML-Based Sensitivity-Aware Selective Image Encryption Algorithm." ResearchGate Publication, 2025.
- [4] Smith, J., & Lee, D. "Network Security Through Combined Cryptography and Steganography: A Review Article." MINAR International Journal, 2026.

Thank You...!!